

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 12: Reflection

Step-by-step solutions to the previous year questions

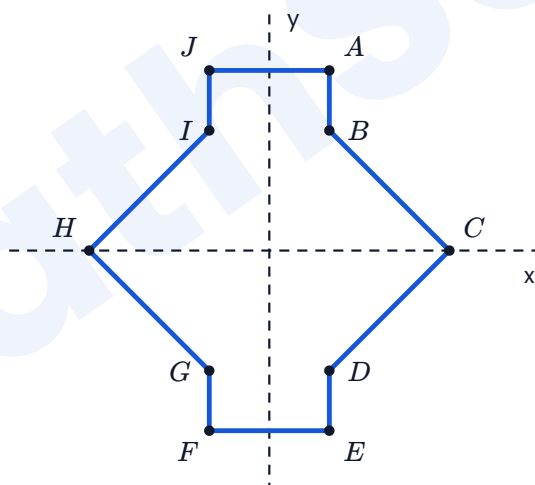
The question numbers match the PYQ question sheet.

1 Reflection of figures

1. Plot $A(1, 3)$, $B(1, 2)$ and $C(3, 0)$ on graph paper. (a) Reflect A and B in the x -axis to get E and D . (b) Reflect A and B in the origin to get F and G . (c) Reflect A , B and C in the y -axis to get J , I and H . (d) Join $A, B, C, D, E, F, G, H, I, J$ in order and name the closed figure. **ICSE 2025, 5 marks**

SOLUTION

- (a) Reflection in the x -axis changes the sign of y : $E(1, -3)$ and $D(1, -2)$.
 (b) Reflection in the origin changes both signs: $F(-1, -3)$ and $G(-1, -2)$.
 (c) Reflection in the y -axis changes the sign of x : $J(-1, 3)$, $I(-1, 2)$ and $H(-3, 0)$.



- (d) The figure has 10 sides and two of its angles point inwards on each side.

Answer: a concave decagon, shaped like a double-headed arrow

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